Thesis

Me, Myself, I

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1 Overview

This software allows you to explore human anatomy in 3D. Find anatomy parts in the tree hierarchy and draw them. Create arbitrary cross sections and challenge your knowledge with automatically generated quizzes.
The program is split into two parts: a tree hierarchy on the left and a view for 3D visualization on the right. The tree hierarchy organizes human anatomy by easy to follow concepts and allows you to find what you are looking for quickly. Advanced users may also modify the hierarchy or use the program to visualize their own custom hierarchy and files.
The 3D view on the right can be used to visualize items from the tree and create arbitrary cross sections. Finally, the quiz mode provides an opportunity to challenge yourself.

2 Working with the Tree

2.1 Selection

Select items via mouse click, or hold down **Shift** or **Ctrl** and click to select multiple items.
The toolbar above the tree provides useful utility functions that apply to all items you have selected:

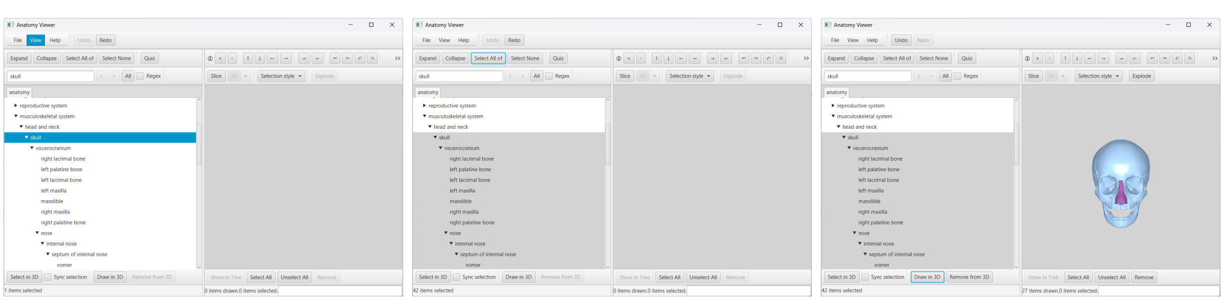
- **Expand** Expands all items below
- **Collapse** Collapses all items below
- **Select All of** Selects all items below
- **Select None** Clears the tree selection
- **Expand** Expands all items below

Use the search field (**Ctrl** + **F**) to search for items.

2.2 Drawing stuff

The toolbar below the tree allows you to add and remove items from the 3D view.

- **Draw in 3D** draws the items you've selected if they are associated with any 3D files. ?? shows an example of how to draw the skull.
- **Remove from 3D** removes the selected items from the 3D view.



(a) Find the skull (b) Select all children of the skull (c) Draw selected items

Figure 2: Example of how to draw something from the tree

2.3 Interacting with the 3D view

Move or rotate the 3D view via the buttons above the 3D view or by using mouse and keyboard:

- **Click** to select items
- **Drag** to rotate

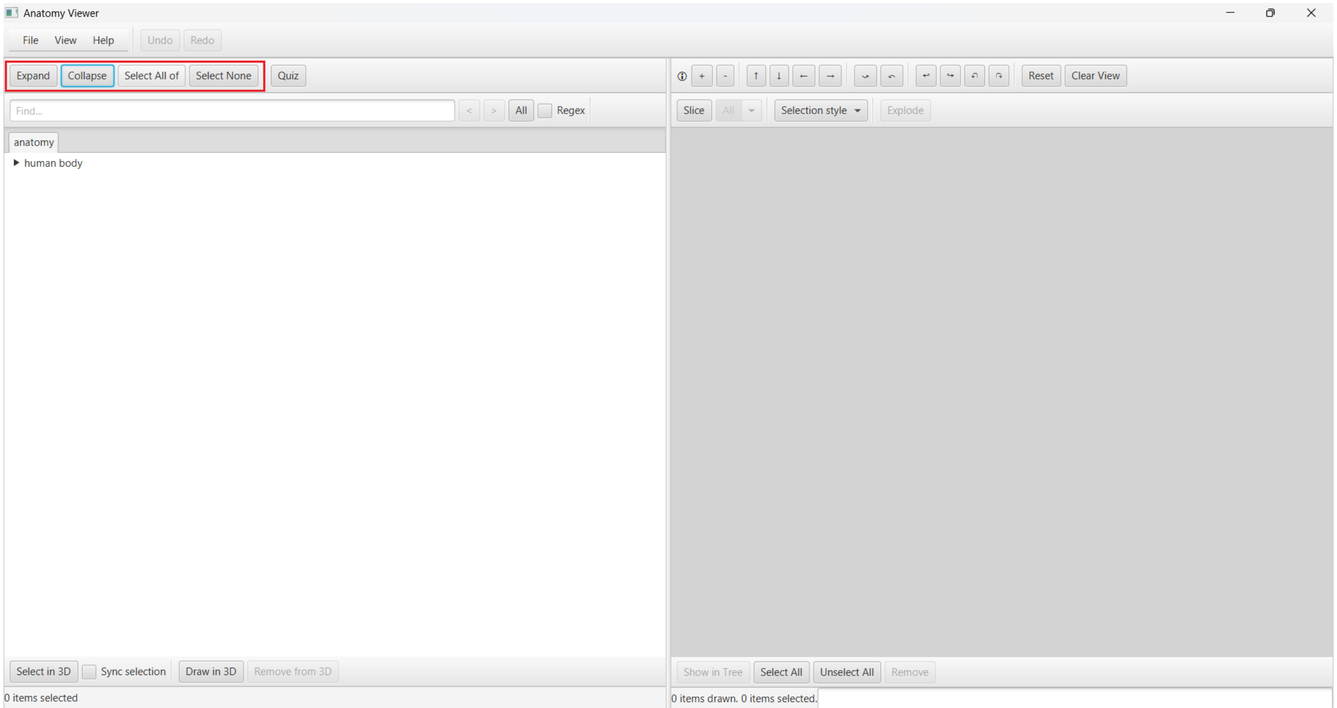


Figure 1: Buttons to modify the selection of the tree

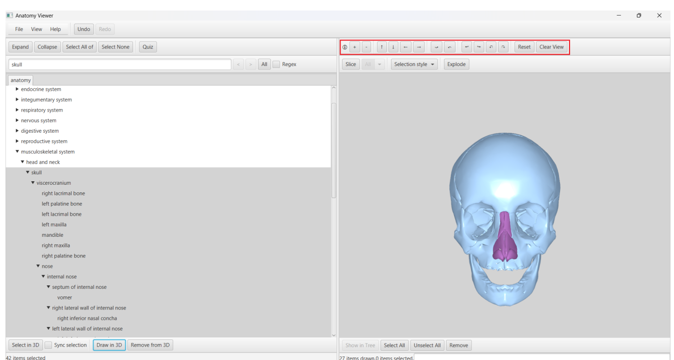


Figure 3: Button controls for the 3D view

  - **Ctrl+Shift+Drag** to trigger continuous rotation
- **Scroll** to zoom
  - **Ctrl+Scroll** to move vertically
  - **Alt+Scroll** to move horizontally
- **Ctrl+Arrow keys / +- keys** to move
- **Alt+Arrow keys / +- keys** to rotate

If a slice plane is drawn, hold down shift while performing any of these actions to instead move the slice plane.**??**

2.3.1 Resetting the 3D view

To return to the default point of view, press **Reset View**. To also clear the 3D view of all drawn items, press **Clear view**.

2.4 Selecting 3D objects

Much like the nodes in the tree, 3D objects can also be selected. Do so by clicking on them. This will cause the selected selection style to be applied to them. By default, selected items will be colored more intensely. The selection style can be changed via the dropdown menu above the 3D view4.
Click **Remove** to undraw all selected items.
Click **Select All** or **Unselect All** select or unselect all items. To highlight the items you've selected in 3D in the tree, click **Show in tree**.
The inverse is also possible. To highlight a specific node from the tree in the 3D view, click **Select in 3D** below the tree.

2.4.1 Synchronizing selection between Tree and 3D view

By default, selecting a node in the tree will not also select it in the 3D view. By checking **Sync selection**5, items that are selected in 3D will also be selected in the tree and vice versa if they are drawn.

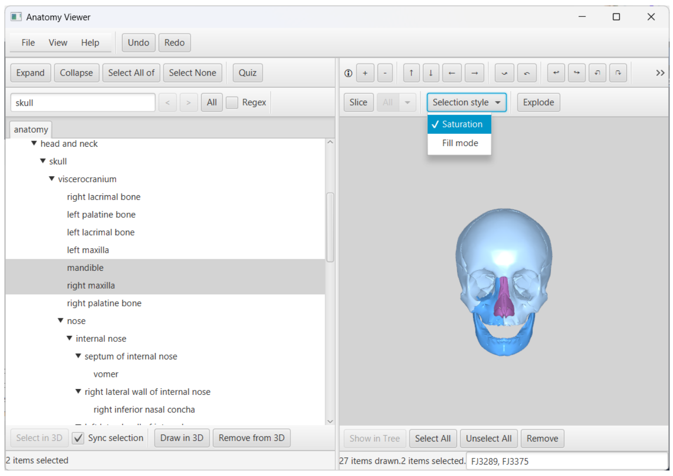


Figure 4: Change the selection style

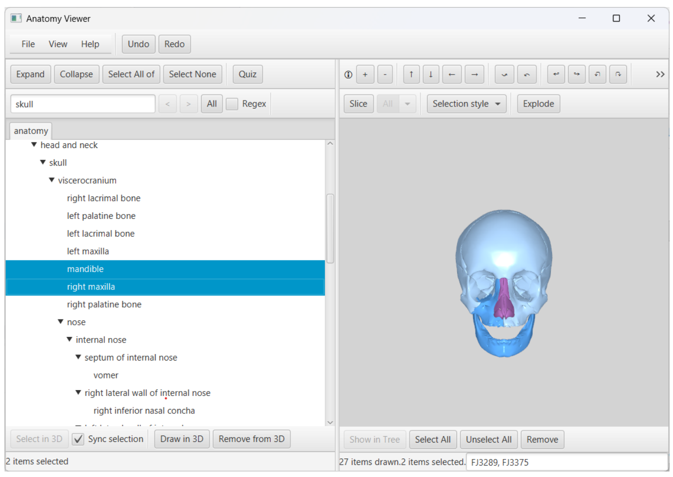


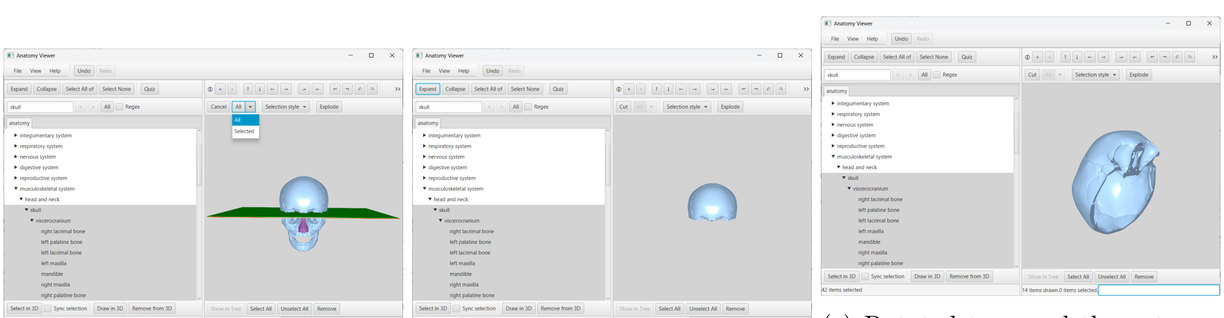
Figure 5: Caption

2.5 Tree Data Structure

2.6 Cross sections / Slicing

Pressing **Slice** above the 3D view causes a red and green plane to appear (only if something is drawn). Next to the button is a dropdown button with default set to 'All'. Pressing **All** will slice through all 3D objects along that plane. What lies on the red side will be cut off while what lies on the green side will be kept.
By choosing **Selected** instead of **All**, only selected objects will be sliced.

To change the orientation of the slice plane, use the same mouse and keyboard controls as for moving the 3D objects2.3, but hold down **Shift** while doing so.

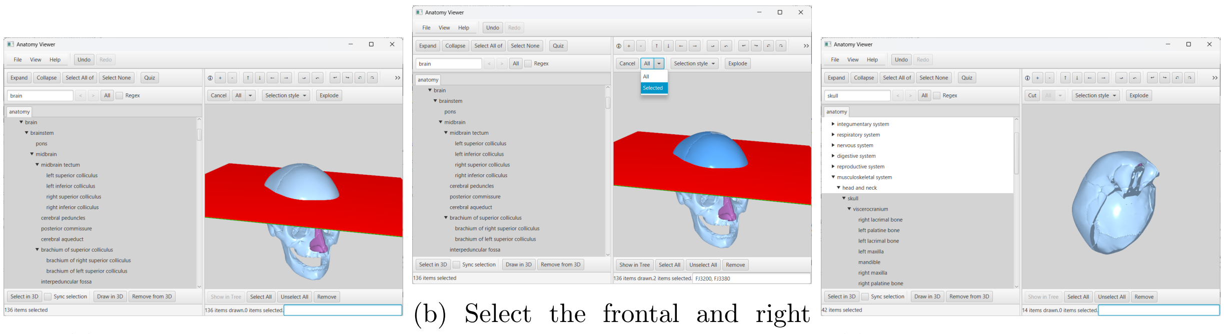


(a) Slice all drawn objects

(b) The skull has been sliced

(c) Rotated to reveal the cut surface

Figure 6: Slicing example



(a) Move the slice plane (b) Select the frontal and right parietal bone (c) Slice only the selected items

Figure 7: Slicing example 2. This time also the brain has been drawn. Only the frontal and right parietal bone are selected and cut.

3 Data model

The underlying data model is a tree in which each node has:

- a unique ID ("conceptID")
- a possibly redundant name
- a possibly empty list of file IDs

3.1 Loading custom models

File > **Load hierarchy** allows you to load a custom model. To do so, you must provide at least an edge list representing the tree. Optionally, you may provide a mapping of nodes to files and a directory containing .obj files.
In the edge list, there must be one entry per line and each entry must be tab separated and adhere to the following format:

```
parent id	parent name	child id	child name
```

`child id` and `child name` may be omitted.
Note that:

- if the provided edge list has more than one possible root, all roots will be made children of a new root
- any node may only have up to one parent
- cycles will be removed
- the first line of the list is treated as header and will be ignored

In the mapping of nodes to files, there must be one entry per line and each entry must be tab separated and adhere to the following format:

```
node id	node name	file id
```

Note that:

- file id is the name of an .obj file
- the first line of the list is treated as header and will be ignored

Note: Adding custom models is undoable via the **Undo** button. It is also possible to remove custom trees by right-clicking their tab and choosing **Remove**.

3.1.1 Generating the edge list and file list for the tree

To generate the edge list for any subtree, right click on a node and press **Print subtree**. This will generate the edge list for the subtree below that node. Click **Print FileIDs of subtree** to generate the mapping of nodes to File IDs for the subtree below that node.

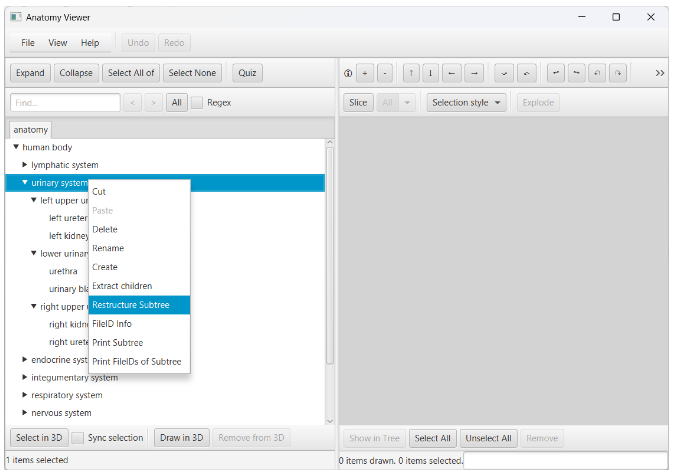


Figure 8: The **Restructure subtree** menu

3.2 FileID Info

To inspect which 3D files are associated with any node in the tree, **right-click** > **FileID Info**. This brings up a two-column table.
Cells in the left column contain a file ID. For every file ID, the right column lists all the nodes that have that file ID. In total, the table lists all file IDs of the subtree below the node that you are inspecting.
The right column is color coded to the type of items it references. If a cell references the item you are inspecting, it is colored light green. For other items, it is colored beige if it is a leaf node (i.e. does not have any children) and dark green if it is an internal node (i.e. has children).
Click the column headers to sort the table alphabetically.
Enter a File ID in the bottom text field and hit search to scroll to it in the table.
If tree editing 3.3is enabled, you may add or remove File IDs from this node by entering them in the text field and pressing **Add FileID** / **Remove FileID**.
The text field in the bottom right corner contains the name and concept ID of this node.

3.3 Tree Editing

The program allows you to edit the tree. Enable **File** > **Enable Tree Editing**.
This will enable the **Add FileID** and **Remove FileID** options in 3.2.
Also, the context menu for tree items will be expanded by the following options:

- **Cut**: cut the item. Not available for the root
- **Paste**: paste the item
- **Delete**: delete the item. Not available for the root
- **Rename**: rename the item
- **Create**: create a new child below that item
- **Extract children**: makes all the children of that node siblings of it. Not available for the root
- **Restructure subtree**: allows you to replace the node and its subtree with a new tree by providing a new edge list. See 3.3.2

3.3.1 Saving changes

To save changes to a tree, right click on the tab of that tree and choose **Save**. The program will ask you to choose a directory where the two files (edge list and node to File ID mapping) will be generated.
Note: This means that changes to the main model ('anatomy') can only be saved by saving them as a new model. They will **NOT** be loaded again upon startup of the program. You must load the modified model via **File** > **Load hierarchy**.
To save all trees, regardless of whether they contain unsaved changes, use **File** > **Save trees**.
If there are unsaved changes in any tree, the program will ask you to save them before exiting or before removing the tree.

3.3.2 Restructure subtree

Opening the **Restructure subtree** function on a node allows you to replace the node and the subtree below it by a new tree.
In the top text field, paste an edge list (format described in 3, NO HEADER LINE) and click **Parse**. This will create a new tree from the provided edge list. Nodes that already existed in the tree will be reused

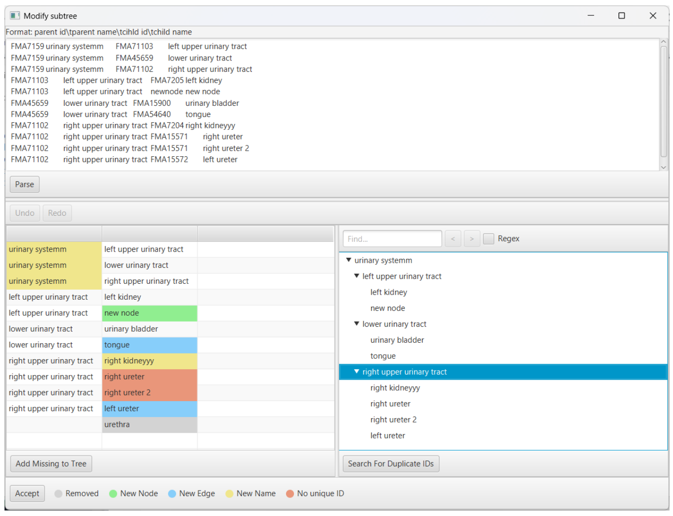


Figure 9: Using the **Restructure subtree** function to restructure the subtree of 'urinary system'8.

(meaning they will also keep their fileIDs). The tree will appear in the bottom right window. The tree can be edited.
In the bottom left window, there is a table giving an overview over the changes this new tree has compared to the old one. Each row is an edge (parent left, child right). The table is color-coded according to the legend at the bottom.
If the new tree does not contain all items of the old subtree, this will be indicated in the table. By clicking **Add missing items to tree**, these items will be added to the tree as children of the root.
Click **Accept** in the bottom left corner to replace the node with which you opened **Restructure subtree** by the new tree.
Since each node must have a unique conceptID (see 3), no two nodes in the new tree must share the same conceptID. Further, nodes of the new tree must not have conceptIDs that already occur somewhere else in the tree outside of the subtree you are restructuring. The program will check both upon clicking **Accept**. The former can also be checked via **Search For Duplicate IDs**. If duplicate conceptIDs are found, a table is shown showing which new item shares a conceptID with which other or already existing item.

Fig.9 shows how the **Restructure subtree** function can be used. As an example, a new tree with a few nonsensical modifications has been parsed:

- 'urinary system' has been renamed to 'urinary systemm'
- a new node 'new node' has been created below 'left upper urinary tract'
- the 'tongue' node has been copied from elsewhere in the tree to below 'lower urinary tract'
- 'right kidney' has been renamed to 'right kidneyyy'
- a new node 'right ureter 2' has been created under 'right upper urinary tract'. It has the same conceptID as 'right ureter'
- the 'left ureter' has been moved to below 'right upper urinary tract'
- 'urethra' has been removed

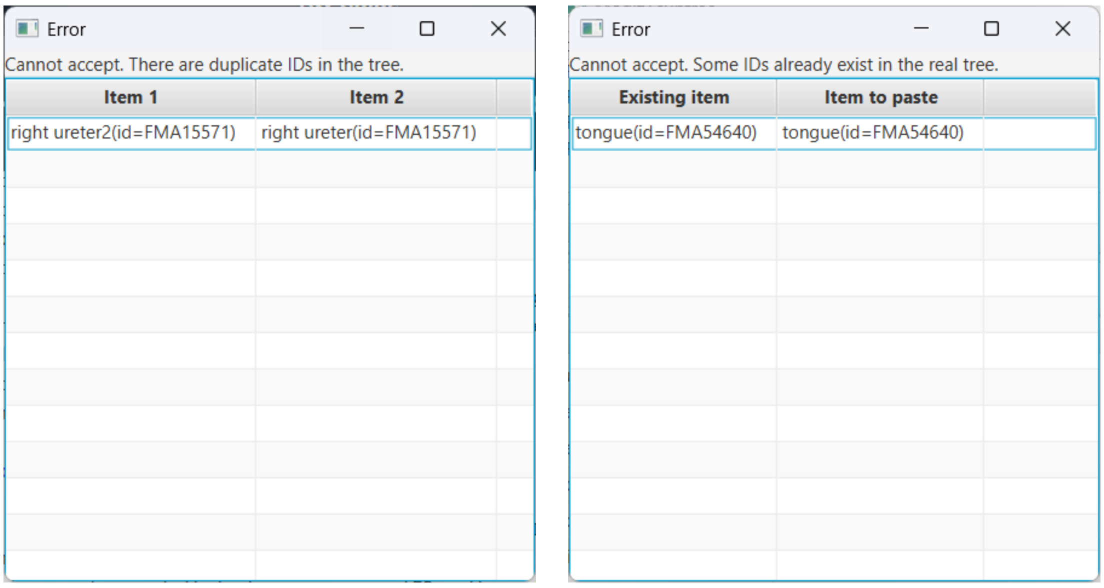
Trying to **Accept** this tree will not work because:

- there are two nodes in it that share the same conceptID, namely 'right ureter' and 'right ureter 2'
- there is a node in it that shares a conceptID with another node elsewhere in the tree, namely the 'tongue'. This is only checked after all duplicate IDs within the new tree have been resolved.

The program notifies the user of these problems (Fig.10).

4 BodyParts 3D 3.0

The BodyParts3D database comes in different versions. The versions share most anatomy parts, but there are some anatomy parts that can only be found in one version.
By default, this program uses version 4.0. By enabling version 3.0 (**File** > **Enable BP3D 3.0**), items labeled with '(available in BP3D 3.0)' will become drawable. Further, you will be able to browse the full tree of version 3.0.
Note that versions 4.0 and 3.0 do not share the same coordinates and sizes! This means that even identical anatomy parts from different versions will be offset from each other. Items from version 3.0 will have a slightly wrong location when drawn next to items from version 4.0. The severity of this offset is different



(a) Two nodes share the same fileID. This is not allowed.

(b) The 'tongue's fileID already exists elsewhere in the tree. This is not allowed.

Figure 10: Problems with the restructured tree of 'urinary system'.

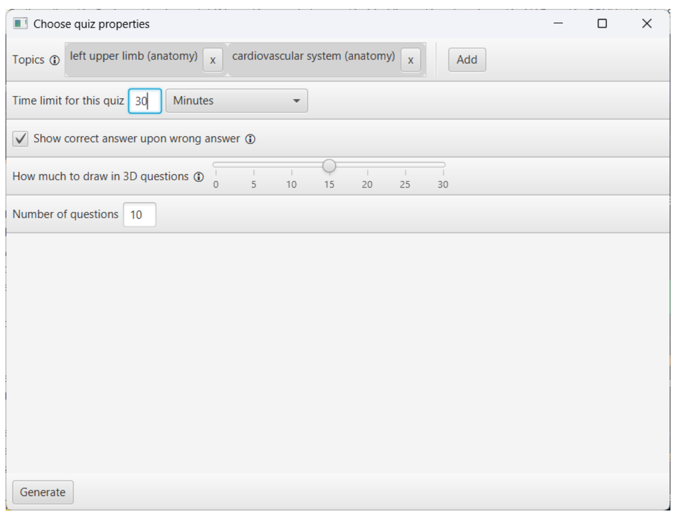


Figure 11: Setting up a quiz

from part to part. You can see this, for example, by drawing the tibia of versions 4.0 and 3.0 next to each other.
Version 3.0 will always be drawn in default green to always make them distinguishable.

5 Quiz

Press the **Quiz** button to start the quiz setup. There you can choose parameters based on which the program will create a quiz. You can choose:

- the topic(s). A topic is represented by a node from a tree. The quiz will ask about items in the subtree of the chosen node(s)
- the time limit for the quiz
- whether the quiz will show you the correct answer if you submit a wrong answer
- how many items will the quiz draw in 3D questions
- the number of questions in the quiz (max. 30)

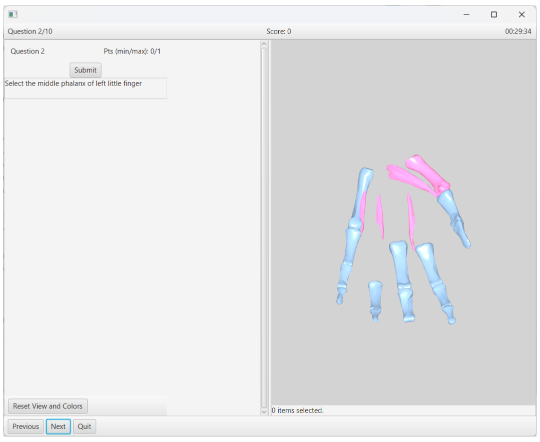
After choosing at least one topic, click **Generate** to start the quiz.

5.1 Question types

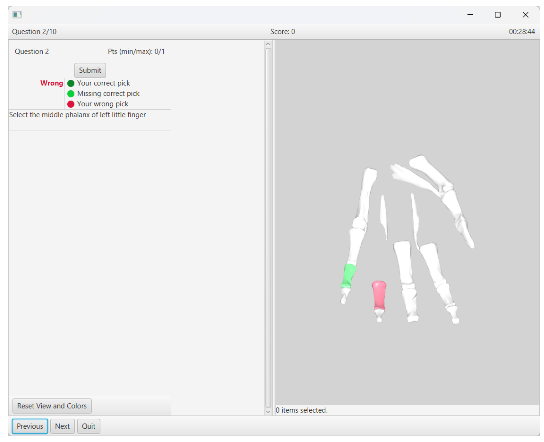
There are four general types of questions:

- *Select in 3D question* (Fig.12): a few things are drawn in 3D and you are asked to select one of them
- *Multiple choice 3D question* (see Fig.13, 14): a few things are draw in 3D and one of them is selected/highlighted. A multiple choice questions asks the name of the selected item

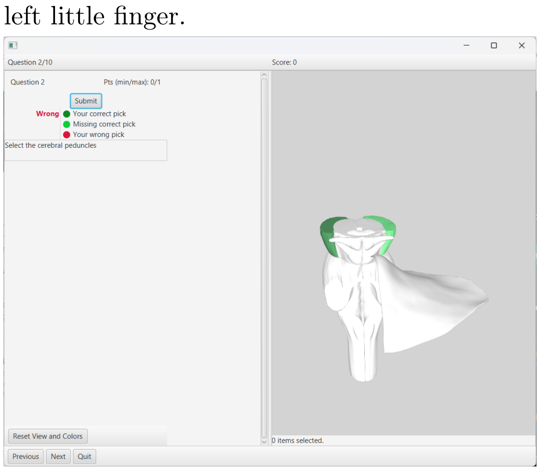
- *Multiple choice Cross section question* (see Fig.15): a cross section of the full human body is shown. One item is highlighted and a multiple choice question asks the name of the highlighted item.
- *Blood vessel continuity question* (see Fig.16): Based on the hierarchy in the tree, the question asks about the direct parent/children of a blood vessel. Note: the hierarchy of blood vessels in the tree is based on vessel continuity, i.e. which vessel feeds into / branches off of which. However, there may be errors or misleading structure that may cause faulty or misleading questions.



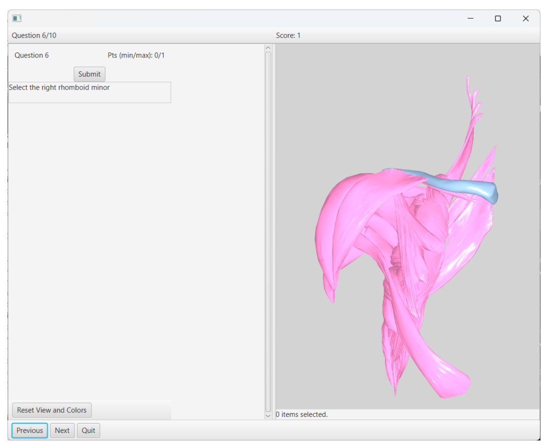
(a) A few bones and muscles are shown. The user is asked to select the middle phalanx of left little finger.



(b) The user failed to correctly pick the middle phalanx of left littel finger. The user's wrong choice is highlighted in red while the correct choice is highlighted in green.

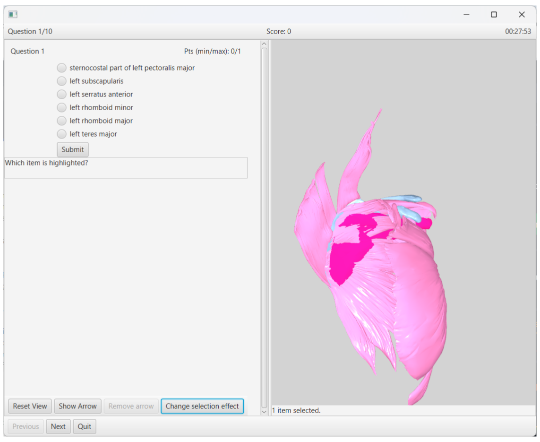


(c) The user was supposed to pick the cerebral peduncles, but only picked one of them. Their correct choice is highlighted in dark green while the missing peduncle is highlighted in light green.

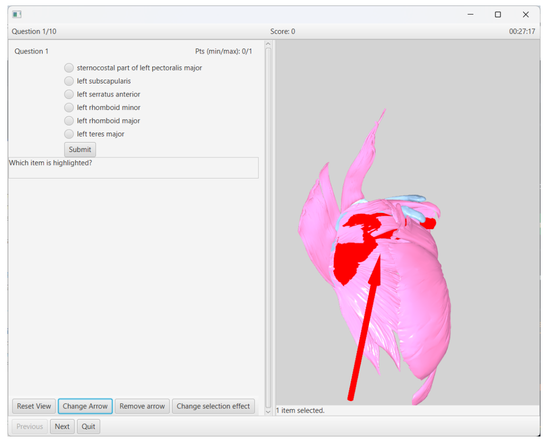


(d) Some muscles and bones of the shoulder are drawn and the user is asked to select the right rhomboid minor.

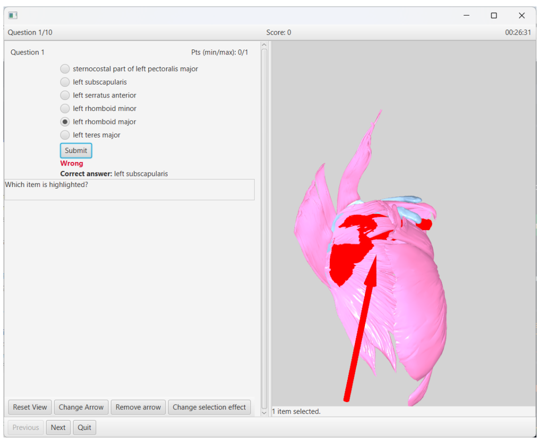
Figure 12: Examples for the 'select in 3D' question type.



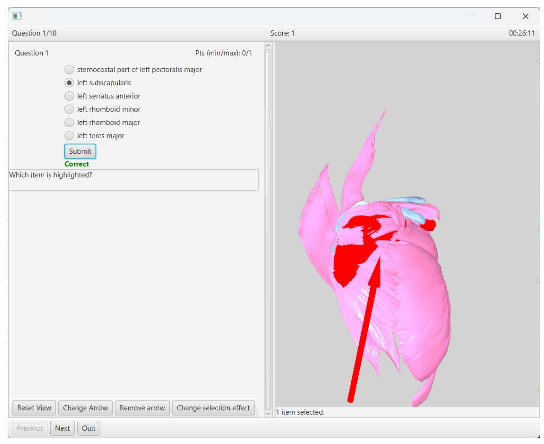
(a) Multiple muscles and bones are drawn and one of them is highlighted. A multiple choice question asks the identity of the highlighted structure.



(b) Clicking **Show Arrow** highlights the structure in red and points an arrow at it.

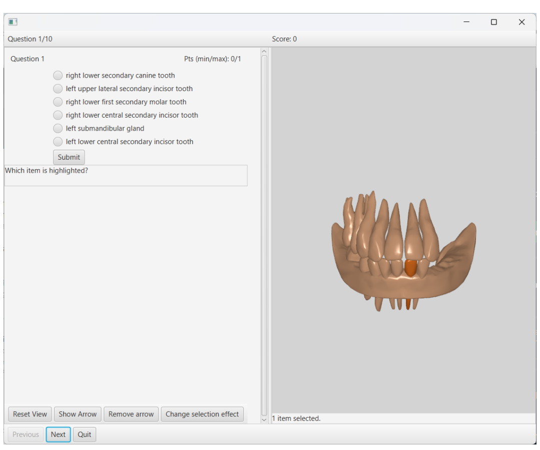


(c) The user wrongly answers 'left rhomboid major' upon which the corret answer 'left subscapularis' is revealed.

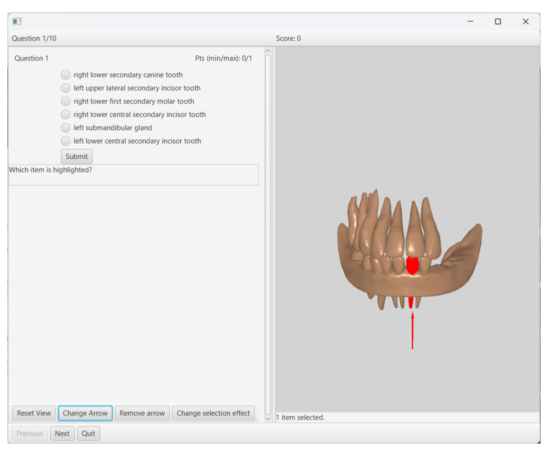


(d) The user chooses the correct answer.

Figure 13: An example of how a multiple choice 3D question.

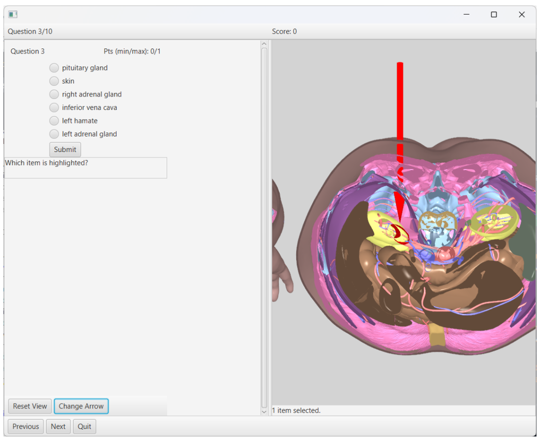


(a) Multiple teeth are drawn and one of them is selected. A multiple choice question asks the identity of the selected item.

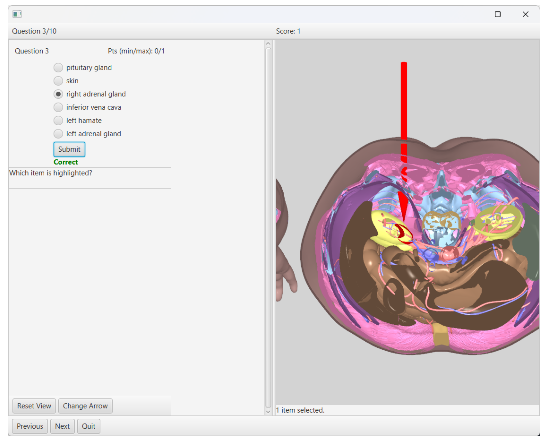


(b) The **Show Arrow** function highlights the item in red and points an arrow at it.

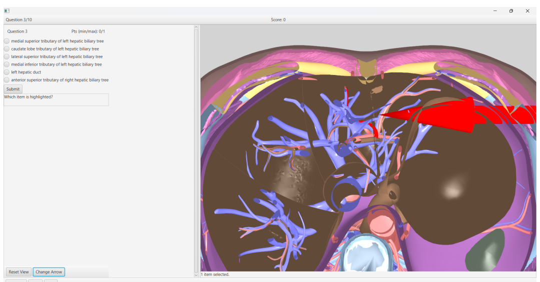
Figure 14: Another example for a multiple choice 3D question.



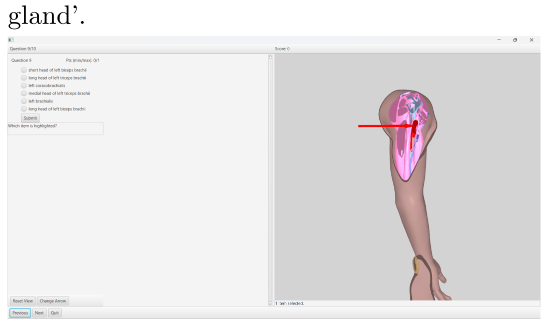
(a) A cross section through the abdominal area is shown from above. An arrow points at an item highlighted in red. A multiple choice question asks the identity of the highlighted item.



(b) The user correctly answers 'right adrenal gland'.



(c) A cross section through the liver is shown. An arrow points at a structure highlighted in red and a multiple choice question asks the identity of the structure.



(d) A slice through the upper arm is shown. An arrow points at a structure highlighted in red and a multiple choice question asks the identity of the structure.

Figure 15: Examples for cross section questions.

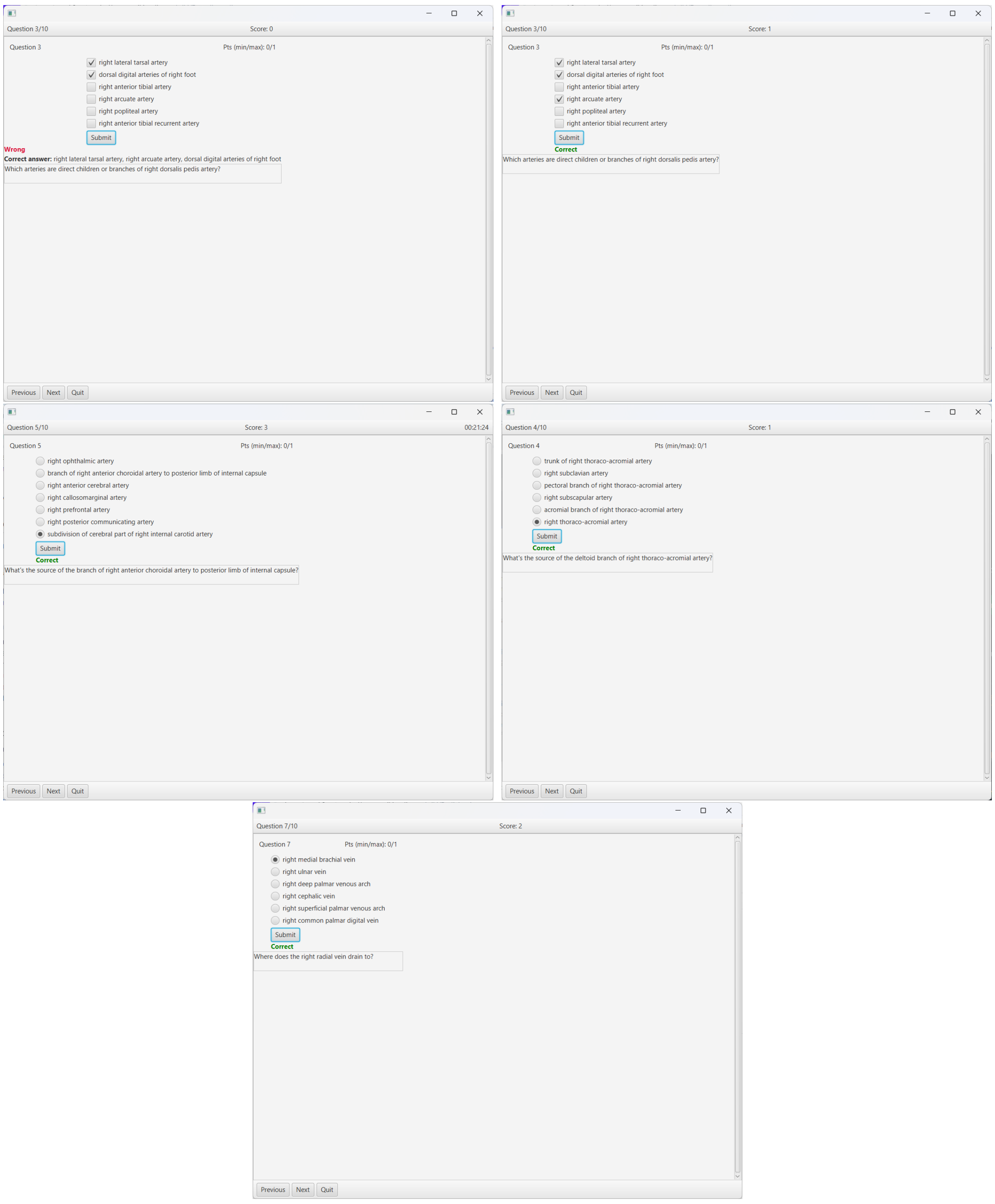


Figure 16: Examples for questions asking about blood vessel continuity.

6 Help

Go to **Help** > **Guide** to show this guide.